

Stagehand Phase Evidence

Current checkpoint: public Site v69, verified 2026-09-03.

Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: github.com/swaymun/stagehand-animation-studio

Commit: `b4e95cf02541837594486481ce18e9fac6452042`

Public Site: stagehand-animation-studio.saimun-h-shahee.chatgpt.site

Sites version: 69

Acceptance loop

Implement - local tests - Sites deployment - hosted test - UI roast - fixes - second review.

Implemented at this checkpoint

- Revision-safe WebMCP mutations with optional expectedRevision and structured REVISION_CONFLICT results.
- Idempotent WebMCP mutations with optional idempotencyKey replay protection.
- Storyboard mode selects the Board rail; Preview selects Scenes and hides editing mutation surfaces.
- Preview canvas suppresses selection chrome and remains aligned with the deterministic render evaluator.
- Character transform inputs have explicit accessible names and the smoke gate checks numeric controls.

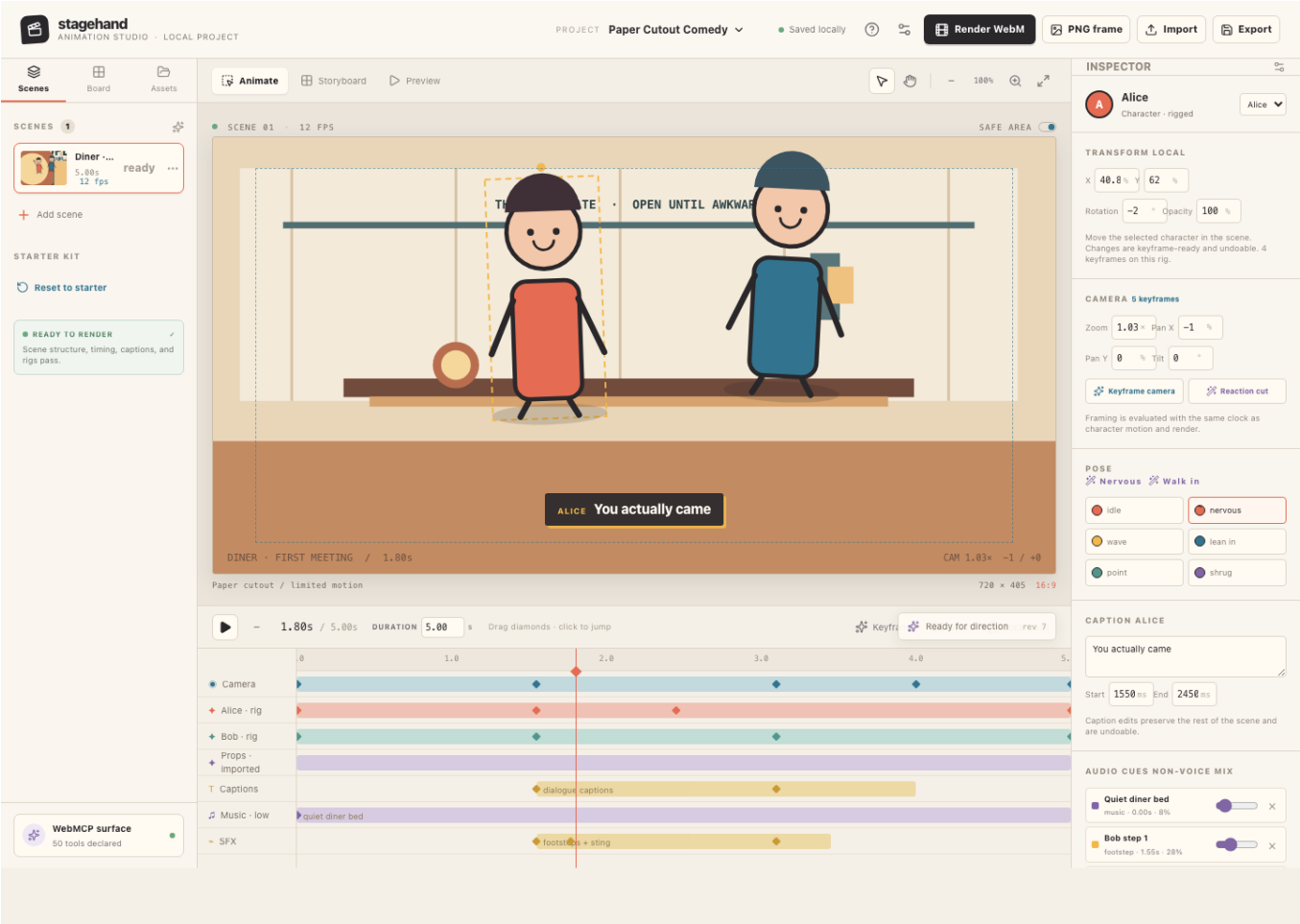
Verification

Local gate:

```
npm run format
npm run lint
npm run build
npm run smoke
```

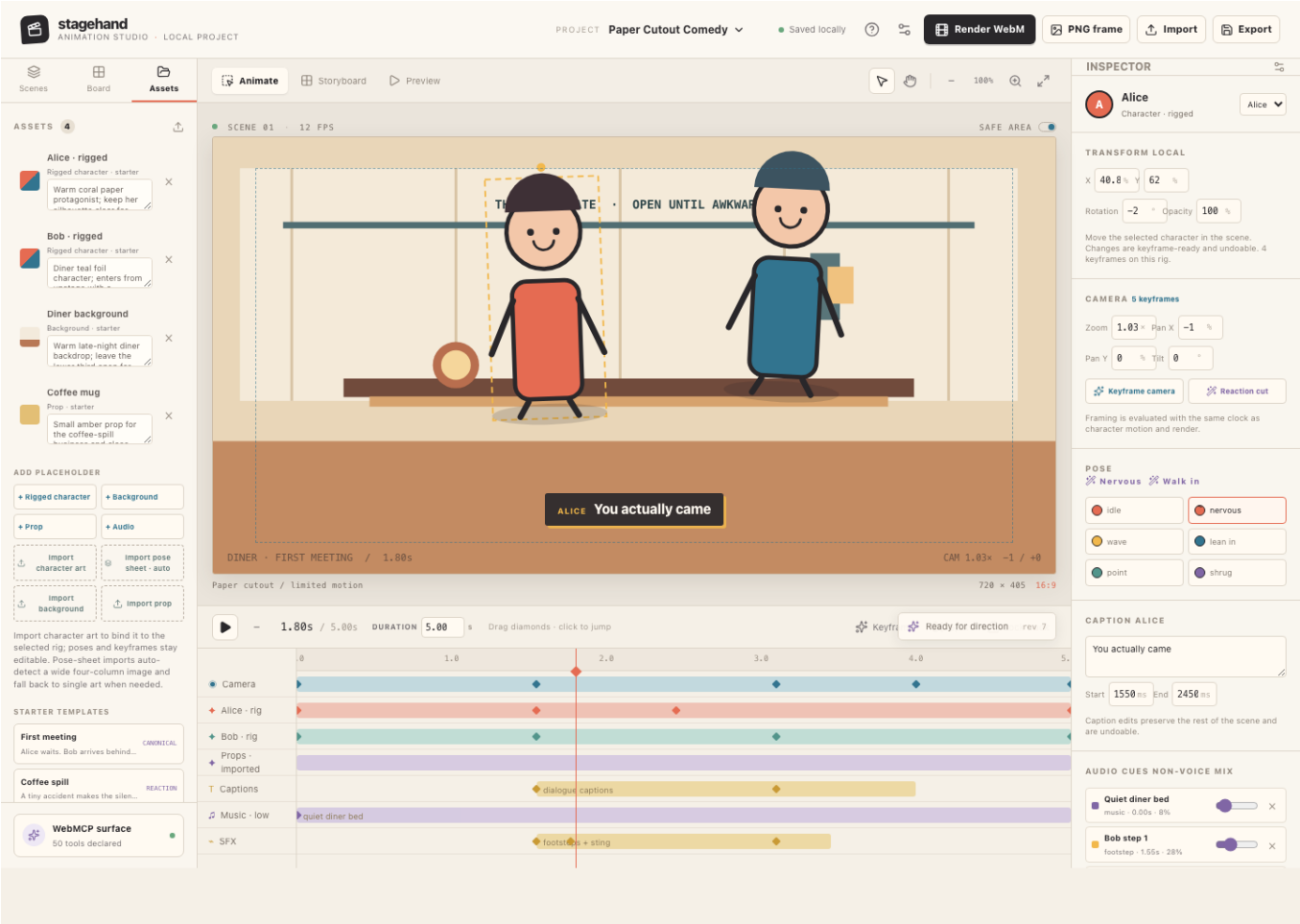
Local and hosted smoke both passed with 50 registered tools, 37 guarded mutations, stale-write conflict, idempotent replay, timeline retiming, imported-prop animation, undo/redo, a two-scene WebM download, synchronized Storyboard/Preview state, and zero page errors.

Animate editing workspace



Hosted Playwright capture at 1440 x 960. The small black fixture in the Assets flow is intentionally a test import.

Assets library and imported-prop controls



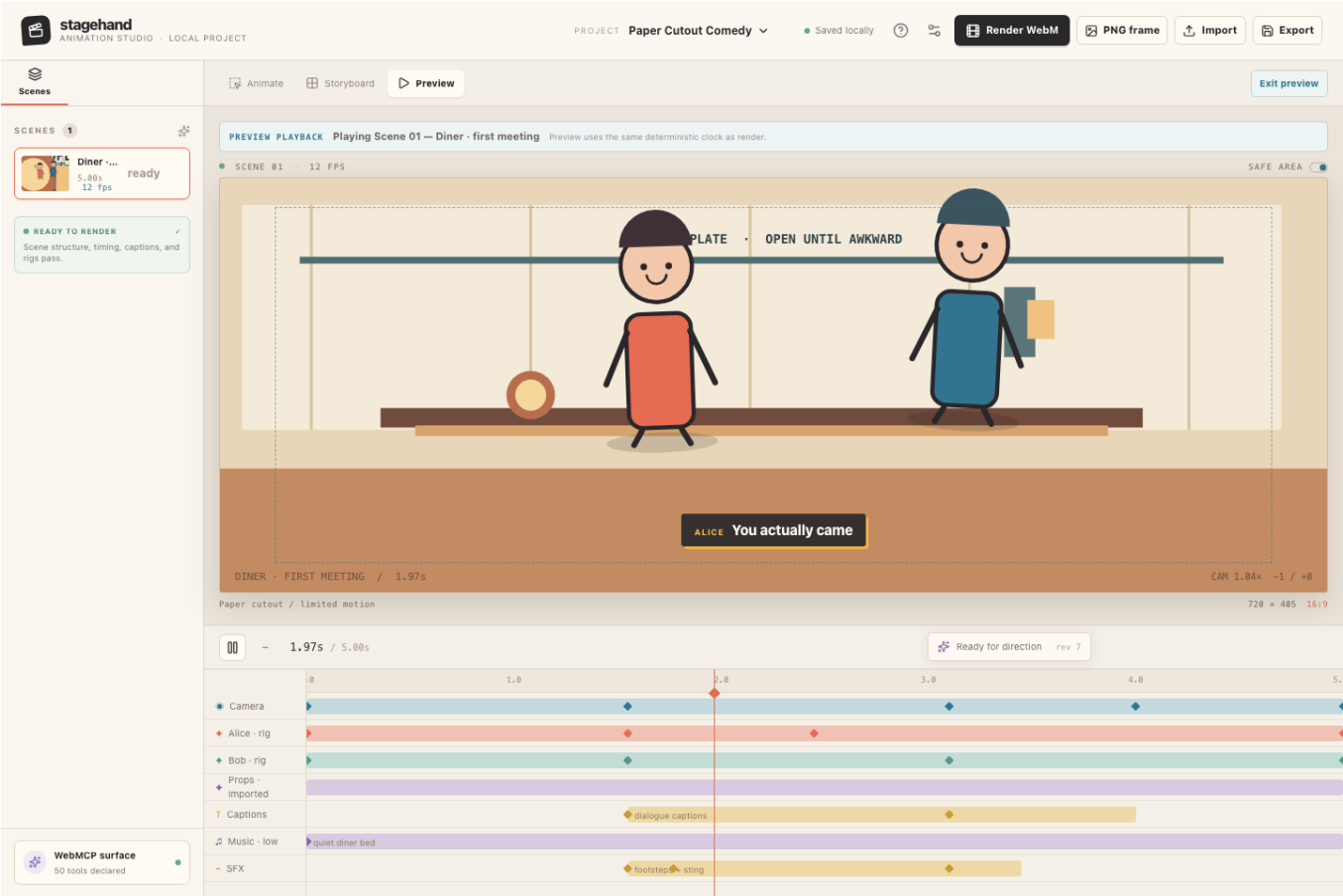
Hosted Playwright capture at 1440 x 960. The small black fixture in the Assets flow is intentionally a test import.

Storyboard mode with Board rail selected

The screenshot shows the Stagehand Animation Studio interface in Storyboard mode. The top bar includes the project name "Paper Cutout Comedy", a "Render WebM" button, and options for "PNG frame", "Import", and "Export". The left sidebar shows the "Board" tab selected, with a list of beats: "01 The wait", "02 The entrance", and "03 The pause". The main workspace displays a storyboard with three panels. The first panel is titled "The wait" and shows Alice practicing what to say. The second panel is titled "The entrance" and shows Bob arriving behind her. The third panel is titled "The pause" and shows neither knowing what to do. Below the storyboard is a timeline with a duration of 5.00s. The timeline shows keyframes for Camera, Alice - rig, Bob - rig, Props - Imported, Captions, Music - low, and SFX. The right sidebar shows the "Inspector" panel for the selected character, Alice. It includes a "Transform Local" section with position, rotation, and opacity controls. The "Camera" section shows keyframes for zoom, pan, and tilt. The "Pose" section shows a list of poses: "Nervous" (selected), "Walk in", "Idle", "Wave", "Lean in", "Point", and "Shrug". The "Caption Alice" section shows the text "You actually came" and a start/end time range. The "Audio Cues Non-Voice Mix" section shows a list of audio cues: "Quiet diner bed" and "Bob step 1".

Hosted Playwright capture at 1440 x 960. The small black fixture in the Assets flow is intentionally a test import.

Review-first Preview player



Hosted Playwright capture at 1440 x 960. The small black fixture in the Assets flow is intentionally a test import.

Roast findings and fixes

- Imported props lacked direct human transform editing. Added X/Y/Scale/Rot controls sharing the keyframe command path.
- Prop fields initially truncated. Replaced the inherited flex behavior with a readable two-column grid.
- Preview carried too much editing chrome. Converted it into a review-first player with scene-only context and Exit preview.
- Timeline diamonds had tiny hit targets. Enlarged the buttons to 22 x 22 pixels without changing the visual diamond.
- Agent retries could duplicate successful writes. Added idempotency-key replay protection.
- Switching Storyboard mode could leave Assets selected. Synced mode selection to the companion project rail.
- Character transform inputs lacked explicit accessible names. Added dynamic labels and a smoke check.

Limits

Direct CUA inspection was unavailable because the Mac was locked; Playwright was used for hosted interaction and screenshots. The smoke harness proves injected modelContext registration and execution, not live production ChatGPT WebMCP discovery. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.